

1. Introduction

Althöfer, Haugland, Scherer, Schneider and Van Cleemput [1] introduced alternating plane graphs: plane graphs in which adjacent vertices have different degrees and adjacent faces have different sizes. The condition is severe, and it makes even existence questions hard. Their paper closes ([1], §10, p. 362) with four open problems, of which the first is:

Conjecture 10.1. There are no 2, Y -alternating plane graphs and no X , 2-alternating plane graphs.

Here an X , Y -alternating plane graph has exactly X distinct vertex degrees and exactly Y distinct face sizes. We prove the conjecture (Theorem 4.1).

The proof is short, and its engine is [1]’s own. Its §9.1 bounds the face count of a *weak* alternating plane graph with degrees 2 and k , by summing $1/r + 1/s$ over the edges separating an r -face from an s -face. That derivation never uses the degree 2. Meanwhile §3.2 attacks the strong X , Y class with different machinery, obtaining only the lower bounds $V \geq 25$ and $V \geq 56$ and never bounding the face count. Connecting the two is what settles the conjecture.

Disclosure of AI use Large language models were used in this work and the extent is material. The step of §5 that removes the (C2) hypothesis was proposed by an AI reviewer; the author re-derived it independently and every bound in it is checked here. AI review also found errors in earlier drafts: a false justification inside Step 0 of Theorem 5.3, and two false explanatory remarks elsewhere. All are corrected here. The manuscript was drafted with AI assistance. The author has verified its mathematical content and is answerable for it and for this disclosure; the prose is not claimed to be the author’s own sentence by sentence. No AI system is an author, and responsibility for every claim rests with the named author.

2. Preliminaries

We follow [1] throughout.

DEFINITION 2.1 ([1], Definition 2.1). An *alternating plane graph* is a simple connected plane graph in which

- (a) every vertex has degree at least 3;
- (b) every face, the unbounded one included, has size at least 3;
- (c) adjacent vertices have different degrees;
- (d) adjacent faces have different sizes.

DEFINITION 2.2 ([1], Definitions 3.1 and 3.4). A $(3, 4, 5)$ -alternating plane graph is one in which every vertex degree and every face size lies in $\{3, 4, 5\}$. An X , Y -alternating plane graph is one with exactly X distinct vertex degrees and exactly Y distinct face sizes.

CONVENTION 2.3 (face size). The *size* of a face is the length of its boundary walk counted with multiplicity, that is, the number of edge-side incidences. It is not the number of distinct vertices on the face. The two differ exactly when the walk repeats a vertex, which is a property of one face and not of the graph. In particular a bridge contributes 2 to the size of the single face carrying it.

[1] does not separately define “size”. This is nonetheless the reading its own arguments require, in three independent places. Equations (3.1) and (9.2) both assert $\sum_s s f_s = 2e$, which counts edge-side incidences. The proof of Lemma 9.2 states that, the graph being bipartite, $f_j = 0$ for every odd j — valid for boundary-walk length, and *not* in general for the number of distinct boundary vertices, since a bipartite facial walk through a cut vertex can meet an odd number of them. And p. 339 treats a bridge as producing a face adjacent to itself, so the paper’s face language is not confined to simple cycles. We use boundary-walk length throughout.

Condition (d) of Definition 2.1 does not mention bridges, and at a bridge one face lies on both sides of an edge, so the condition is either violated there or vacuous there. Two readings are therefore available in principle, but [1] does choose between them, as we record next.

CONVENTION 2.4 (bridges). Say that a plane graph satisfies (C2) if no edge has the same face on both sides, equivalently if it has no bridge. [1] takes (d) to include a face adjacent to itself, so that (C2) follows from Definition 2.1; this is stated explicitly on p. 339, where an alternating plane graph is said to be “always at least 2-edge-connected, since plane graph with edge connectivity 1 contains a face that is adjacent to itself”. Under the *weak reading*, (d) constrains only pairs of distinct faces and bridges are permitted.

The proof of Conjecture 10.1 in §4 uses (C2), and we prove in §5 that a putative counterexample must be bridgeless under *either* reading. The result is therefore independent of the choice. We stress that this is a robustness statement and not the repair of a gap: the authors of [1] state the strict reading, and we simply decline to rely on it.

We shall use the following, which is due to [1].

3. The per-edge identity

The following double count is the engine of §4. It is the convention behind (9.2) of [1], isolated.

LEMMA 3.1. *Let G be a finite plane graph with no isolated vertex, with v vertices, e edges and f faces. For an edge a with ends u, w and face occurrences $F^-(a)$, $F^+(a)$ on its two sides, put*

$$c(a) = \frac{1}{\deg u} + \frac{1}{\deg w} + \frac{1}{|F^-(a)|} + \frac{1}{|F^+(a)|}.$$

Then $\sum_a c(a) = v + f$. If G is connected, Euler's formula gives

$$\sum_a c(a) = e + 2, \quad \text{equivalently} \quad \sum_a (c(a) - 1) = 2.$$

PROOF. A vertex of degree d is an end of exactly d edges and contributes $1/d$ each time, hence 1 in total; summing gives v . By Convention 2.3 a face of size s occurs exactly s times as a side of an edge and contributes $1/s$ each time, hence 1 in total; summing gives f . $\square$

REMARK 3.2. Neither count requires facial walks to be simple. Repeated vertices, cut vertices and bridges are all permitted; a bridge simply has $F^-(a) = F^+(a)$ and contributes $2/|F|$ from its face side. This is what makes Lemma 3.1 usable in §5, where bridges are the object of study.

We write $x(a) = c(a) - 1$ for the *excess* of an edge, so that Lemma 3.1 reads $\sum_a x(a) = 2$: in a connected plane graph some edge must have positive excess.

4. Conjecture 10.1

THEOREM 4.1. *There is no 2, Y -alternating plane graph and no X , 2-alternating plane graph.*

We prove the two halves separately, then remove the dependence on Convention 2.4 in §5. Both halves are the same inequality with the roles of vertices and faces exchanged; no dual graph is constructed, so neither invokes the 3-edge-connectivity hypothesis of ([1], Lemma 2.2).

4.1. No 2, Y -alternating plane graph Suppose G is one, with degrees taking exactly the two values $d_1 < d_2$.

Adjacent vertices differ in degree and only two degrees occur, so every edge joins a d_1 -vertex to a d_2 -vertex. In particular G is bipartite, and every edge contributes

$$\frac{1}{d_1} + \frac{1}{d_2} \leq \frac{1}{3} + \frac{1}{4} = \frac{7}{12}$$

to the vertex side, using $d_1 \geq 3$ from Definition 2.1(a).

Since G is bipartite, every closed walk has even length, so by Convention 2.3 every face has even size, hence size at least 4. By (C2) the two sides of an edge are distinct faces, and by Definition 2.1(d) their sizes differ; being distinct even numbers at least 4, one is at least 4 and the other at least 6. So the face side contributes at most $\frac{1}{4} + \frac{1}{6} = \frac{5}{12}$.

Hence $c(a) \leq \frac{7}{12} + \frac{5}{12} = 1$ for every edge, so $\sum_a x(a) \leq 0$, contradicting Lemma 3.1. $\square$

REMARK 4.2. This is (9.5) of [1] with $1/2$ replaced by $1/d_1 \leq 1/3$. The derivation of (9.5) never used the degree 2; §9.1 of [1] runs the chain on the weak class with degrees 2 and k , while §3.2 attacks the strong class with different machinery yielding only $V \geq 25$ and $V \geq 56$ and never bounding f . The two sections were not connected.

4.2. No $X, 2$ -alternating plane graph Suppose G is one, with face sizes taking exactly the two values $s_1 < s_2$.

By (C2) the two sides of every edge are distinct faces, and by Definition 2.1(d) they have different sizes; with only two sizes available, every edge separates an s_1 -face from an s_2 -face and contributes at most $\frac{1}{3} + \frac{1}{4} = \frac{7}{12}$ to the face side.

We claim every degree is even. Fix a vertex u of degree d , let its edges in rotation order be $a_1, \dots, a_d$, and let F_i be the face occupying the corner between a_i and a_{i+1} , indices modulo d . The two sides of a_i are the occurrences F_{i-1} and F_i , distinct faces by (C2), so $|F_{i-1}| \neq |F_i|$. With only two sizes the cyclic word $|F_1|, \dots, |F_d|$ alternates, and a cyclically alternating binary word has even length. Hence d is even.

Stating the argument over corners rather than over “the incident faces” is what makes it correct at a cut vertex, where one face occupies several corners: those corners are non-consecutive, so alternation is undisturbed.

Degrees are at least 3 and even, hence at least 4; adjacent vertices have distinct even degrees, so the vertex side contributes at most $\frac{1}{4} + \frac{1}{6} = \frac{5}{12}$. Again $c(a) \leq 1$ for every edge and Lemma 3.1 is contradicted. $\square$

REMARK 4.3. *Euler is used as $v - e + f = 2$; alternating plane graphs are connected by Definition 2.1, and disconnectedness would only strengthen the contradiction.*

4.3. Calibration Run on the weak $2, k$ class that [1] settles, the chain excludes exactly $k \geq 12$, reproducing ([1], (9.10)), and does not exclude $k \leq 10$, where that paper exhibits weak $2, k$ -alternating plane graphs; $k = 11$ it leaves open, as does [1] without its Lemma 9.4. So the contradiction turns on $d_1 \geq 3$ and not on an over-count, which would also have killed the small- k graphs.

This is emphatically *not* a certification. A spuriously *stronger* face bound such as $5/12 - 1/1000$ reproduces the same threshold exactly, so the calibration cannot detect an invalid strengthening — the dangerous direction. It rules out one failure mode and no others.

5. Removing the dependence on (C2)

Everything in this section is under the weak reading of Convention 2.4: condition (d) of Definition 2.1 constrains only pairs of distinct faces, and bridges are permitted. We show that no $X, 2$ -alternating plane graph exists even so, and hence that Theorem 4.1 holds under both readings. Note the scope: this does *not* say that every weak-reading alternating plane graph is bridgeless. It is not — join three copies of a $(3, 4, 5)$ -alternating plane graph by two bridges drawn in the exterior, at a degree-5 vertex of each. The joined degrees become 7, 6, 6, their old neighbours keep degree 3 or 4, the merged exterior face has size at least 13 and so differs from every interior face, and the result is a weak-reading alternating plane graph with bridges. It has neither exactly two degrees nor exactly two face sizes, so it is no counterexample to Theorem 4.1. What is proved here is bridgelessness *within the two-face-size class*. The corresponding statement for $2, Y$ is Proposition 5.5 at the end of the section, proved by the same per-edge count.

The weak reading introduces exactly one new difficulty: a bridge need not force a change of face size, so a vertex on a bridge need not have even degree. The lemma below controls that defect, and the proof then runs as a charge. If a bridge exists, the face carrying it has size at least 8, which makes every bridge contribute excess at most $-\frac{1}{6}$, while the only edges that can have *positive* excess are non-bridge 3–4 edges, each at most $\frac{1}{24}$. The lemma attaches every positive edge to a unique bridge at its degree-3 end, and each bridge receives at most two. The bridges’ deficit then dominates every positive excess, contradicting $\sum_a x(a) = 2$. If there is no bridge, the even-degree argument of §4.2 applies directly.

LEMMA 5.1 (parity). *Let G be a plane graph whose faces admit a colouring by two colours that is proper on every pair of distinct adjacent faces. Then for every vertex v ,*

$$\deg v - |\{\text{bridges at } v\}| \text{ is even.}$$

PROOF. Walk the darts at v in rotation order and let $F_1, \dots, F_d$ be the corner faces as in §4.2, so that F_{i-1} and F_i are the two sides of the edge a_i . These two occurrences are of the same face precisely when a_i is a bridge. Hence the corner colour changes across each non-bridge edge at v and is unchanged across each bridge at v . The walk closes up, so the number of changes is even, and that number is $\deg v$ minus the number of bridges at v . $\square$

Lemma 5.1 is the bridge-tolerant replacement for the classical Eulerian criterion, and it is strictly weaker than that criterion rather than a restatement of it: a triangle with a pendant edge has its two faces properly coloured once self-adjacency across the bridge is ignored, yet is not Eulerian. With no bridges the lemma gives even degree at every vertex — the direction of the usual Eulerian criterion needed here, and the only one it supplies. In an $X, 2$ -alternating plane graph the face size supplies the colouring, so the lemma applies.

COROLLARY 5.2. *In an $X, 2$ -alternating plane graph under the weak reading, a vertex lying on no bridge has even degree, hence degree at least 4.*

THEOREM 5.3. *Under the weak reading there is no $X, 2$ -alternating plane graph. In particular no such graph has a bridge.*

PROOF. Let G be one, with face sizes $s_1 < s_2$, and let B be its number of bridges. Suppose first $B \geq 1$.

Step 0: $s_2 \geq 8$. Let $a = uw$ be a bridge, and let A and C be the components of $G - a$ containing u and w . Then $\deg_A u = \deg u - 1 \geq 2$ and likewise $\deg_C w \geq 2$, so both have at least three vertices. Let W_A be the facial boundary walk of A through the corner exposed by deleting a . Its length is not 1, which needs a loop. Nor 2: that is either a digon of parallel edges, excluded by simplicity, or a single edge traversed in both directions, and for those two darts to form the whole facial boundary both endpoints of that edge have degree 1, forcing $A = K_2$ against $\deg_A u \geq 2$. So $|W_A| \geq 3$, and

likewise $|W_C| \geq 3$. The face F of G incident with both sides of a is traversed as W_A , then a , then W_C , then a in the opposite direction, so by Convention 2.3

$$|F| = |W_A| + |W_C| + 2 \geq 3 + 3 + 2 = 8.$$

As $|F| \in \{s_1, s_2\}$ and $s_1 < s_2$, we get $s_2 \geq 8$.

Step 1: only an edge at a vertex of degree 3 can have positive excess. Recall $s_1 \geq 3$ and, by Step 0, $s_2 \geq 8$; degrees at the two ends of an edge are distinct by Definition 2.1(c). Three cases:

- a a bridge: both sides carry F , so $c(a) \leq \frac{1}{3} + \frac{1}{4} + \frac{2}{8} = \frac{5}{6}$ and $x(a) \leq -\frac{1}{6}$;
 - a not a bridge, both ends of degree at least 4: $c(a) \leq \frac{1}{4} + \frac{1}{5} + \frac{1}{3} + \frac{1}{8} = \frac{109}{120}$ and $x(a) \leq -\frac{11}{120}$;
 - a not a bridge with an end of degree 3: $c(a) \leq \frac{1}{3} + \frac{1}{4} + \frac{1}{3} + \frac{1}{8} = \frac{25}{24}$ and $x(a) \leq \frac{1}{24}$.
- Only the third class can be positive. Note in particular that a bridge endpoint of degree 5 cannot make an edge positive.

Step 2: each bridge carries at most two positive edges. First, a positive edge is a 3–4 edge. Its face side is at most $\frac{1}{3} + \frac{1}{8} = \frac{11}{24}$, so its vertex side must exceed $\frac{13}{24}$; among pairs of distinct degrees at least 3, only $\frac{1}{3} + \frac{1}{4} = \frac{14}{24}$ does, since already $\frac{1}{3} + \frac{1}{5} = \frac{12.8}{24}$ falls short.

Let a be an edge of positive excess and let u be its end of degree 3, which is unique since adjacent vertices have different degrees. By Lemma 5.1 the number of bridges at u has the same parity as $\deg u = 3$, so it is 1 or 3; it is not 3, since a itself is a non-bridge edge at u . So u lies on exactly one bridge. Write $b(u)$ for it and *charge* a to $b(u)$: every positive edge thus has one, and only one, destination.

The qualification matters: a vertex of degree 3 need not carry two non-bridge edges, since one with three bridges carries none. What is proved, and what is needed, is that a vertex of degree 3 *incident with a non-bridge edge* lies on exactly one bridge and two non-bridge edges.

Now fix a bridge b and count the positive edges charged to it. Each such edge has its degree-3 end at a vertex whose unique bridge is b , so that vertex is an end of b . The two ends of b have different degrees by Definition 2.1(c), so at most one of them has degree 3; call it u if it exists. Of the three edges at u , one is b itself, leaving exactly two that could be positive and charged to b . So each bridge receives at most two charges, and writing P for the number of positive edges, $P \leq 2B$.

Step 3. Partition the edges into the P positive ones, the B bridges, and the rest. By Steps 1 and 2 their excesses are at most $\frac{1}{24}$, $-\frac{1}{6}$ and 0 respectively, so by Lemma 3.1,

$$2 = \sum_a x(a) \leq \frac{P}{24} - \frac{B}{6} \leq \frac{2B}{24} - \frac{B}{6} = -\frac{B}{12} < 0,$$

a contradiction.

Finally suppose $B = 0$. By Corollary 5.2 every degree is even and at least 4, so the vertex side of every edge is at most $\frac{1}{4} + \frac{1}{6} = \frac{5}{12}$; the face side is at most $\frac{1}{3} + \frac{1}{4} = \frac{7}{12}$; hence $x(a) \leq 0$ for every edge and $\sum_a x(a) = 2$ is again contradicted. $\square$

REMARK 5.4 (how much room the argument has). *The tether of Step 2 is the substance: treating the positive edges as unconstrained gives only a lower bound on B and closes nothing. It does not have to be sharp. Ignoring the degree condition on a bridge's two ends gives only $P \leq 4B$, and even then the bound of Step 3 is 0, which still contradicts 2; the sharp tether merely supplies the margin $-B/12$. That margin also means Step 0 is not needed at full strength: with $s_2 \geq 7$ the per-bridge total is $-\frac{1}{84}$, still negative. Some lower bound on the bridge face is nevertheless indispensable — at $s_2 \geq 6$ the total is $+\frac{1}{12}$ — and at $P \leq 5B$ the argument fails outright.*

PROPOSITION 5.5. *Under the weak reading there is no 2, Y -alternating plane graph either.*

PROOF. Step 0 of Theorem 5.3 uses only minimum degree 3 and simplicity, so it applies here: a bridge's face has size at least 8. As in §4 the graph is bipartite, since two degrees and Definition 2.1(c) force every edge to join a d_1 -vertex to a d_2 -vertex; hence every closed walk has even length, a facial walk traversing a bridge twice included, and every face has even size at least 4. Then a bridge has

$$c(a) \leq \frac{1}{3} + \frac{1}{4} + \frac{2}{8} = \frac{5}{6},$$

and a non-bridge edge, whose two sides are distinct faces of distinct even sizes at least 4 and 6, has $c(a) \leq \frac{1}{3} + \frac{1}{4} + \frac{1}{4} + \frac{1}{6} = 1$. So $x(a) \leq 0$ for every edge, contradicting $\sum_a x(a) = 2$. $\square$

References

- [1] I. Althöfer, J. K. Haugland, K. Scherer, F. Schneider and N. Van Cleemput, *Alternating plane graphs*, *Ars Math. Contemp.* **8** (2015), 337–363. doi:10.26493/1855-3974.584.09a.

Hanyu Yang, Independent researcher, California, USA
e-mail: hanyu.yang.92@gmail.com